ENTERPRISE AI VALIDATION SERIES

AI Evaluation Metrics Guide

A practical, trainer-ready reference for validating GenAI applications, RAG pipelines, and agentic AI systems — covering quality, factuality, safety, compliance, and operational readiness.

GenAI | RAG | Agentic AI | Banking & Enterprise Use Cases

MODULE 01

GenAI Evaluation Metrics

Accuracy, relevance, factuality, safety & task-specific validation.

MODULE 02

RAG Evaluation Metrics

Retrieval quality, groundedness, citation accuracy & hallucination control.

MODULE 03

Agentic AI Evaluation Metrics

Planning, tool use, trajectory, workflow compliance & auditability.

01

MODULE ONE

GenAI Evaluation Metrics

A simple validation guide for LLM applications such as chatbots, summarization, classification, report generation, email drafting, and enterprise productivity use cases.

1. Most Important GenAI Metrics to Remember

For most GenAI applications, start with these core metrics. They are easy to explain and useful across banking, HR, operations, analytics, customer support, and knowledge-work use cases.

Metric	Simple Meaning	Why It Matters
Accuracy	Is the output correct?	Prevents wrong business decisions.
Relevance	Does the output answer the user request?	Avoids off-topic or generic answers.
Completeness	Does the output cover all required points?	Prevents missing important information.
Factual Correctness	Are the facts true and verifiable?	Reduces false or misleading content.
Instruction Following	Did the model follow the prompt rules?	Important for enterprise workflows and automation.
Format Compliance	Is the output in the requested format?	Needed for JSON, tables, reports, emails, and APIs.
Safety & Compliance	Is the output safe, private, and policy-compliant?	Critical for regulated domains such as banking.
Hallucination Rate	Did the model invent unsupported information?	Measures one of the biggest GenAI risks.

2. GenAI Metrics with Simple Questions

Use these questions during validation, demo review, UAT, or production monitoring.

Metric	Simple Validation Question
Accuracy	Is the answer or output correct?
Relevance	Does it directly answer the user question?
Completeness	Did it include all required information?
Coherence	Is the answer logically structured and easy to follow?
Fluency	Is the language clear, professional, and readable?
Factual Correctness	Can the facts be verified from a trusted source or reference answer?
Hallucination Rate	Did the model add anything that was not true or not supported?
Instruction Following	Did it follow all rules given in the prompt?
Format Compliance	Did it return the required structure, such as JSON, table, or email?
Consistency	Does it give stable answers for similar inputs?
Bias / Fairness	Does it avoid unfair or discriminatory outputs?
Privacy	Does it avoid exposing personal or confidential information?
Toxicity / Safety	Does it avoid harmful, abusive, or unsafe content?
Latency	Does it respond within acceptable time?
Cost	Is the token / API cost acceptable for business usage?

3. Simple GenAI Evaluation Flow

A good GenAI validation process should check quality, factuality, safety, and operational readiness.

1. **Define Use Case** — Clarify what the GenAI app should do.
   Summarization, classification, report writing, Q&A, email drafting.
2. **Prepare Test Data** — Create realistic input examples.
   Normal cases, edge cases, confusing inputs, unsafe inputs, missing-data cases.
3. **Evaluate Output Quality** — Check usefulness and readability.
   Accuracy, relevance, completeness, coherence, fluency
4. **Evaluate Factuality** — Check whether the output is true.
   Factual correctness, hallucination rate, citation/source validation
5. **Evaluate Prompt Compliance** — Check whether instructions are followed.
   Format compliance, tone, length, JSON schema, required sections
6. **Evaluate Safety** — Check business and policy risks.
   Privacy, toxicity, bias, compliance, confidential data exposure
7. **Evaluate Operations** — Check production readiness.
   Latency, cost, failure rate, escalation rate, user feedback

Simple rule: Do not validate only the final answer. Validate whether the output is useful, true, safe, consistent, and suitable for the business workflow.

4. Explanation of GenAI Evaluation Metrics

Accuracy

Checks whether the GenAI output is correct for the given task. For classification, it means the predicted class is correct. For Q&A, it means the answer matches the expected answer.

> **Example:** If a complaint should be classified as High Priority and the model predicts High Priority, it is accurate.

Relevance

Checks whether the output is related to the user request. Sometimes an answer may be grammatically correct but still not answer the actual question.

> **Example:** If the user asks for key risks in a banking report, the answer should focus on risks, not general banking definitions.

Completeness

Checks whether all required points are included.

> **Example:** A good customer complaint summary may need issue, customer impact, root cause, action taken, pending owner, and SLA status. Missing any major point reduces completeness.

Coherence

Instruction Following

Checks whether the model followed the prompt rules.

> **Example:** If the prompt says "answer in 5 bullet points" or "do not mention customer name," the model should follow those instructions exactly.

Format Compliance

Checks whether the output follows the requested structure. Important when GenAI output is used by another system — valid JSON, a fixed table format, a standard email template, or a required report structure.

Consistency

Checks whether the model gives stable outputs for similar inputs. A production application should not give very different decisions for nearly identical cases unless there is a valid reason.

Bias and Fairness

Checks whether the model treats users, customers, or groups unfairly.

> **Example:** A loan-support assistant should not make biased assumptions based on gender, region, language, or background.

Privacy and Data Leakage

Checks whether the answer is logically organized. The answer should have a clear flow and should not contradict itself. Important for reports, executive summaries, and customer-facing responses.

Fluency

Checks language quality. The output should be readable, professional, and grammatically correct. For enterprise use, fluency matters because poor language can reduce trust.

Factual Correctness

Checks whether the information in the output is true. Usually validated against source documents, a reference answer, expert review, or trusted data. Critical when the output influences decisions.

Hallucination Rate

Measures how often the model invents information.

> **Example:** If a policy does not mention a refund, but the model says the customer will receive a refund, that is hallucination.

Checks whether the model avoids exposing personal, confidential, or sensitive information — account numbers, passwords, private customer details, or internal-only content.

Toxicity and Safety

Checks whether the output avoids harmful, abusive, unsafe, or inappropriate content. Important for public-facing chatbots and internal enterprise assistants.

Latency & Cost per Request

Latency measures how long the system takes to respond — even a high-quality answer may not be acceptable if too slow. Cost per request measures token, API, infrastructure, and tool usage cost, helping decide whether the application can scale economically.

5. Task-Specific GenAI Metrics

Different GenAI use cases need additional task-specific metrics.

Use Case	Important Metrics	Example
Summarization	Coverage, conciseness, factual correctness, no hallucination	Does the summary include all important points without adding false information?
Classification	Accuracy, precision, recall, F1 score, confusion matrix	Did it correctly classify complaints by risk level or category?
Sentiment Analysis	Accuracy, precision, recall, F1 score by class	Did it correctly identify positive, neutral, or negative customer sentiment?
Chatbot	Task completion rate, resolution rate, escalation accuracy, user satisfaction	Did the chatbot solve the user issue or escalate correctly?
Report Generation	Completeness, structure, factual correctness, business relevance	Did the report include findings, risks, actions, and recommendations?
Email Drafting	Tone, clarity, instruction following, privacy, professionalism	Is the email polite, clear, and safe to send?
Code Generation	Correctness, test pass rate, security, maintainability	Does the generated code run and pass test cases?

6. Practical Banking Example

Use case: A GenAI assistant summarizes customer complaints for a banking operations manager.

Metric	Validation Question
Accuracy	Did it correctly identify the complaint issue?
Completeness	Did it include customer impact, product, issue type, SLA status, and pending action?
Factual Correctness	Did it avoid adding details not present in the complaint text?
Instruction Following	Did it follow the required management-summary format?
Privacy	Did it avoid exposing unnecessary customer PII?
Tone	Is the language professional and suitable for management?
Escalation Safety	Did it flag high-risk or urgent issues correctly?

7. Recommended Validation Set

For a practical GenAI application, prepare a test set with different input types: easy cases, complex cases, ambiguous cases, missing-information cases, unsafe requests, privacy-risk cases, and edge cases. Then score each output using the selected metrics.

MINIMUM ENTERPRISE VALIDATION

Accuracy + Relevance + Completeness + Factual Correctness + Instruction Following + Privacy + Safety + Latency + Cost.

02

MODULE TWO

RAG Evaluation Metrics

Validation metrics for Retrieval-Augmented Generation applications — from retrieval quality through generation quality to business readiness.

1. Most Important RAG Metrics to Remember

For practical RAG validation, focus on these six metrics first. They are enough to explain whether a RAG system is retrieving the right knowledge and generating a reliable answer.

Metric	What It Checks	Why It Matters
Context Precision	Whether the retrieved chunks are useful and relevant.	Reduces noise and prevents the LLM from using unrelated information.
Context Recall	Whether all required information was retrieved.	Prevents incomplete answers caused by missing source content.
Faithfulness / Groundedness	Whether the answer is supported by the retrieved context.	Detects hallucination and unsupported claims.
Answer Relevance	Whether the answer directly addresses the user question.	Ensures the response is useful, not generic or off-topic.
Answer Correctness	Whether the final answer is factually correct.	Checks the answer against expert-approved or reference answers.
Citation Accuracy	Whether citations point to the correct document or section.	Supports auditability, verification, and compliance.

2. RAG Metrics with Simple Questions

Useful for trainers, reviewers, business users, and QA teams when validating RAG outputs.

Metric	Simple Validation Question
Context Precision	Are the retrieved chunks useful for this question?
Context Recall	Did the system retrieve all information needed to answer correctly?
Context Relevance	Are the retrieved chunks related to the question?
Faithfulness / Groundedness	Is the answer fully supported by the retrieved context?
Answer Relevance	Does the answer directly answer the user question?
Answer Correctness	Is the answer factually correct compared with the expected answer?
Citation Accuracy	Are the cited sources correct and verifiable?
Hallucination Rate	Did the model invent anything that is not in the source content?
Context Utilization	Did the model actually use the retrieved context properly?
Noise Sensitivity	Can the model ignore irrelevant or outdated chunks?

3. Simple RAG Evaluation Flow

A RAG application should be evaluated in three stages: retrieval, generation, and business readiness.

1. **Retrieval Quality** — Check whether the retriever found the right documents or chunks.
   Context relevance, context precision, context recall, Top-K accuracy, MRR

2. **Generation Quality** — Check whether the LLM generated a useful and grounded answer.
   Faithfulness, answer relevance, answer correctness, completeness, hallucination rate

3. **Business Readiness** — Check whether the system is safe, usable, auditable, and cost-effective.
   Citation accuracy, latency, cost per query, user satisfaction, compliance safety

4. Detailed Explanation of RAG Evaluation Metrics

RAG evaluation means checking whether the system retrieves the right information, uses that information properly, and generates a correct answer grounded in trusted sources.

4.1 Context Precision

Checks whether the retrieved chunks are actually useful for answering the question — out of all retrieved chunks, how many are relevant?

> **Example:** If the user asks about home loan documents and the system retrieves home loan checklist, KYC requirements, and credit card rewards, the credit card rewards chunk is noise.

Low context precision means the LLM receives unnecessary information, which can increase hallucination risk.

4.2 Context Recall

Checks whether the system retrieved all necessary information required to answer the question — did the retriever find everything needed?

> **Example:** If account closure requires customer form, zero balance confirmation, no pending lien, and identity verification, but the system retrieves only two of them, recall is low.

Low context recall causes incomplete answers even when the final response sounds professional.

4.3 Context Relevance

Checks whether the retrieved content is related to the user question.

> **Example:** For a dispute resolution question, relevant chunks include dispute policy, chargeback process, and complaint escalation rules. Irrelevant chunks such as ATM cash loading schedules or employee leave policy should not be retrieved.

This metric is useful for quickly identifying whether retrieval is going in the right direction.

4.4 Faithfulness / Groundedness

Checks whether the generated answer is supported by the retrieved context — did the model answer only from the provided documents?

> **Example:** If the retrieved policy says loan approval takes 5–7 working days, but the answer says approval takes 24 hours, the answer is not faithful.

This is one of the most important RAG metrics because it directly detects hallucination.

4.5 Answer Relevance

Checks whether the final answer directly answers the user question. An answer can be factually correct but still not useful if it does not address the actual question.

> **Example:** If the user asks for minimum salary required for a personal loan, a generic explanation of personal loans is not enough — the answer should directly mention the salary requirement from the retrieved policy.

4.6 Answer Correctness

Checks whether the final answer is factually correct compared with a reference answer or expert-approved answer. Faithfulness checks support from context; correctness checks accuracy against the expected answer.

> **Example:** If the expected answer requires ID proof, address proof, income proof, and completed application form, but the model mentions only ID proof and income proof, the answer is partially correct.

4.7 Citation Accuracy

Checks whether the answer points to the correct source document, page, section, or chunk. Very important for banking, legal, healthcare, insurance, and compliance use cases.

> **Example:** If the answer mentions a late payment fee, the citation should point to the credit card fee policy, not the savings account opening policy.

4.8 Hallucination Rate

Measures how often the system generates information that is not present in the retrieved context. RAG is designed to reduce hallucination by grounding answers in trusted documents.

> **Example:** If the source says account closure takes 3 working days, but the answer adds a cashback reward that is not in the source, that added claim is hallucinated.

4.9 Context Utilization

Checks whether the model actually used the retrieved context while generating the answer. Sometimes retrieval is good, but the model ignores important chunks.

> **Example:** The retriever provides five chunks, but the answer uses only one and misses important information in the others.

This metric helps identify whether the issue is retrieval quality or generation quality.

4.10 Noise Sensitivity

Checks whether the model gets confused when irrelevant, duplicate, or outdated chunks are included in the context.

> **Example:** For an ATM withdrawal limit question, the retrieved context may include the correct policy, an outdated policy, and unrelated marketing text.

A good RAG system should ignore noisy chunks and use the most relevant and current information.

5. Banking Example

Question: What documents are required for opening a current account for a company?

Retrieved context: Company account opening policy · KYC checklist for companies · Board resolution requirement · Credit card rewards document

Generated answer: To open a company current account, the customer must provide company registration documents, KYC documents, authorized signatory details, and board resolution if required.

Metric	Evaluation
Context Precision	Good, but one irrelevant credit card document was retrieved.
Context Recall	Good only if all required account-opening rules were retrieved.
Faithfulness	Good if every answer point is supported by the retrieved company account policy and KYC checklist.
Answer Relevance	Good because the answer directly responds to the document requirement question.
Citation Accuracy	Good only if citations point to the correct company account policy and KYC sections.
Hallucination Rate	Low if the answer does not add unsupported rules, timelines, or charges.

6. Final Summary

RAG evaluation is not just about whether the final answer looks good. A reliable RAG application must retrieve the right content, use it correctly, avoid hallucination, cite the correct sources, and produce an answer that is relevant, complete, and factually correct.

START WITH THESE SIX

Context Precision, Context Recall, Faithfulness, Answer Relevance, Answer Correctness, and Citation Accuracy.

03

MODULE THREE

Agentic AI
Evaluation Metrics

A practical guide to validate AI agents that plan, use tools, call APIs, execute workflows, and produce final outputs.

1. Most Important Agentic AI Metrics to Remember

Agentic AI evaluation is different from normal LLM evaluation. We do not only check the final answer — we also check whether the agent followed the right steps, selected the right tools, passed correct inputs, respected permissions, and created an auditable workflow.

Metric	What It Checks	Simple Example
Task Success Rate	Did the agent complete the user goal?	Generated the correct compliance report.
Tool Selection Accuracy	Did the agent choose the correct tool or API?	Used database search for customer records, not web search.
Parameter Accuracy	Did the agent pass the right inputs to tools?	Correct customer ID, date range, product type.
Trajectory Accuracy	Did the agent follow the correct sequence of steps?	Retrieve data → validate → analyze → draft report.
Workflow Compliance	Did the agent follow business process and policy?	Did not skip KYC or approval checks.
Human-in-the-Loop Accuracy	Did it ask for human approval when needed?	Asked manager approval before final loan decision.
Safety & Permission Compliance	Did it avoid unsafe or unauthorized actions?	Did not send email or update record without permission.
Auditability	Can we review what the agent did and why?	Logs show plan, tool calls, inputs, outputs, and final action.

2. Agentic AI Metrics with Simple Questions

Metric	Simple Validation Question
Task Success Rate	Did the agent finish the task correctly?
Goal Alignment	Did the agent solve the actual user need, not just produce text?
Plan Quality	Was the plan logical, complete, and efficient?
Step Relevance	Were the steps useful, or did the agent do unnecessary work?
Tool Selection Accuracy	Did it choose the correct tool for each step?
Tool Call Success Rate	Did the tool/API calls execute successfully?
Parameter Accuracy	Were the tool inputs correct and complete?

Metric	Simple Validation Question
Error Recovery	Did the agent recover when a tool failed or returned missing data?
Trajectory Accuracy	Did it follow the right path from start to finish?
Workflow Compliance	Did it follow required enterprise/banking process rules?
Human Approval Trigger	Did it request approval for sensitive actions?
Permission Compliance	Did it avoid actions outside user/system permission?
Data Privacy Compliance	Did it protect confidential data and PII?
Auditability	Can we trace decisions, tool calls, and outputs later?

Simple rule: For an agent, a correct final answer is not enough. The path must also be correct, safe, approved, and auditable.

3. Simple Agentic AI Evaluation Flow

1. **Evaluate the user goal** — Did the agent understand the task and expected outcome?
   Goal alignment, task success criteria

2. **Evaluate the plan** — Did it create a logical and complete plan before acting?
   Plan quality, step relevance, missing-step rate

3. **Evaluate tool usage** — Did it choose the right tools and call them correctly?
   Tool selection accuracy, parameter accuracy, tool success rate

4. **Evaluate workflow path** — Did it follow the right sequence and enterprise process?
   Trajectory accuracy, workflow compliance, order correctness

5. **Evaluate final output** — Is the final response/report/action correct and useful?
   Answer correctness, completeness, relevance

6. **Evaluate safety controls** — Did it protect data and avoid unauthorized actions?
   Privacy compliance, permission compliance, human approval accuracy

7. **Evaluate operations** — Is the system reliable, cost-effective, and traceable?
   Latency, cost per task, failure rate, auditability

Banking example: A loan-review agent should not only produce a recommendation. It must check KYC, credit score, income documents, policy rules, fraud signals, exceptions, and human approval rules in the correct order.

4. Detailed Explanation of Agentic AI Metrics

Task Success Rate

Checks whether the agent completed the requested task correctly. For example, if the user asks the agent to prepare a risk summary, the output should contain the correct issue, impact, risk level, recommendation, and next action.

Goal Alignment

Checks whether the agent solved the real business goal. Sometimes an agent gives a polished answer but misses the actual requirement.

> **Example:** The user asks for high-risk cases, but the agent summarizes all cases.

Plan Quality

Checks whether the agent created a good step-by-step plan. A good plan is clear, complete, and not unnecessarily complex. In enterprise use cases, planning is important because actions may affect customers, records, reports, or compliance.

Step Relevance

Checks whether each step taken by the agent was useful. Unnecessary steps increase cost, latency, and risk.

> **Example:** An agent should not call five tools when one internal database query is enough.

Tool Selection Accuracy

Checks whether the agent selected the correct tool. For example, to calculate numbers it should use a calculator or code tool; to read internal policy it should use the document search tool; to create an event it should use the calendar tool.

Parameter Accuracy

Checks whether the agent passed correct values to tools. Even the right tool can fail if the wrong date range, customer ID, file name, branch code, or product category is passed.

Tool Call Success Rate

Measures how many tool calls completed successfully. Failed API calls, bad permissions, missing files, and invalid parameters reduce reliability.

Error Recovery

Checks whether the agent can handle failures. For example, if one data source is unavailable, the agent should retry, use an approved fallback, or clearly explain the limitation instead of inventing information.

Trajectory Accuracy

Trajectory means the full path the agent followed. This metric checks whether the agent followed the right sequence of decisions and tool calls — important because an agent can reach a correct-looking answer through an unsafe or non-compliant path.

Workflow Compliance

Checks whether the agent followed required business rules. In banking, it may need to check KYC, AML, sanctions, credit policy, approval limits, and audit requirements before taking action.

Human-in-the-Loop Accuracy

Checks whether the agent asks for human approval at the right time. Sensitive actions such as approving a loan, sending a customer email, changing a record, or escalating a case should often require approval.

Permission Compliance

Checks whether the agent respects access control and authority. The agent should not read restricted data, modify records, send messages, or trigger workflows unless the user and system permissions allow it.

Data Privacy Compliance

Checks whether the agent protects sensitive information such as account numbers, customer identifiers, financial details, personal data, internal policies, and confidential documents.

Output Correctness

Checks whether the final output is accurate. For example, a case-summary agent should not mix customer names, wrong dates, incorrect account details, or unsupported risk levels.

Completeness

Checks whether the final output includes all required points. A compliance report may need issue, evidence, risk rating, policy reference, recommended action, owner, and deadline.

Latency

Measures how long the agent takes to complete the task. Agents may make multiple tool calls, so latency can become high if planning and tool use are not optimized.

Cost per Task

Measures total cost for one completed task, including LLM tokens, embeddings, database calls, API calls, and tool execution cost.

Auditability

Checks whether the agent keeps a clear record of the user request, plan, tool calls, tool inputs, tool outputs, decisions, approvals, and final response. This is critical for regulated industries.

5. Practical Banking Validation Checklist

Validation Area	Checklist Question
Goal	Did the agent understand the exact banking task?
Data	Did it use the right approved data sources?
Tools	Did it choose the right internal tools/APIs?
Inputs	Did it pass correct parameters such as customer ID, date, branch, product, or case number?
Workflow	Did it follow the correct KYC, AML, risk, approval, and escalation flow?
Safety	Did it avoid exposing PII or confidential data?
Approval	Did it ask for human approval before sensitive actions?
Final Output	Was the answer/report/action correct, complete, and clear?
Audit	Can all steps and decisions be reviewed later?

MOST PRACTICAL TEACHING SUMMARY

Evaluate an AI agent on the final result, the tools it used, the sequence of steps it followed, the business rules it respected, and the audit trail it produced.